

CSE321: COMPUTER NETWORKS // CISCO PACKET TRACER

Mars Habitat Communication Network

A resilient interplanetary network infrastructure connecting Mars habitat zones, research labs, medical bays, and Earth mission control — engineered for security, segmentation, and survival.

TEAM MEMBERS — 02 OPERATORS

MISSION STATUS / • NOMINAL

CSE321: COMPUTER NETWORKS // FINAL BRIEFING

// PROJECT OVERVIEW

Engineering interplanetary connectivity

The Mars Habitat Communication Network simulates a multi-segment Cisco infrastructure spanning the Red Planet's habitat zones and a live Earth uplink — implementing subnetting, VLANs, dynamic routing, NAT, ACLs, and full application-layer services.

06

ROUTERS CONFIGURED

04

HABITAT VLANS SEGMENTED

16+

NETWORK DEVICES LINKED

TEAM // 02

Project Crew

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Both members collaborated equally on all aspects of the project — from network design and routing configurations to security policies, NAT implementation, and end-to-end troubleshooting.

Why Mars needs a **hardened** network

A habitat on another planet introduces constraints no Earth-based network faces. Every design decision answers a real operational threat.

PROBLEM / 01

Strict Bandwidth Constraints

The Mars↔Earth link is high-latency, low-throughput. Internal traffic must never waste interplanetary bandwidth — segmentation is mandatory.

PROBLEM / 02

Multi-Zone Isolation

Research, Medical, Residential, Greenhouse, Robot Maintenance, Tourist, and Security zones must enforce least-privilege communication policies.

PROBLEM / 03

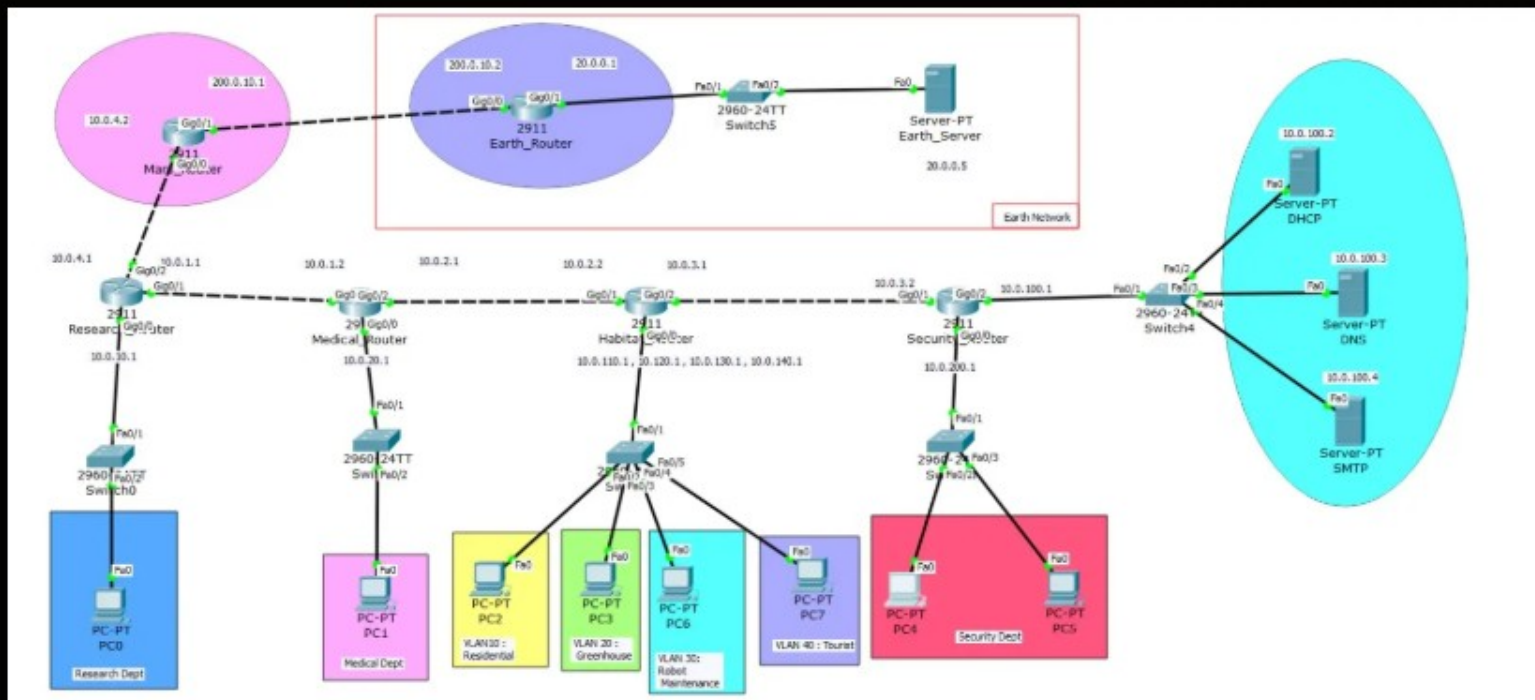
Asymmetric Routing Protocols

EIGRP inside Mars vs OSPF on the Earth uplink requires mutual redistribution without routing loops or black holes.

PROBLEM / 04

Address Translation & Inbound Reachability

Mars internal IPs are private. PAT hides them from Earth — but breaks inbound sessions, requiring deliberate ACL and NAT design.



- ```
[R] RESEARCH
10.0.10.0/24 — Scientific data & analysis

[M] MEDICAL
10.0.20.0/24 — Habitat med bay

[H] HABITAT
4 VLANs — Residential / Greenhouse / Robot / Tourist

[S] SECURITY
10.0.200.0/24 — Surveillance & control

[E] EARTH LINK
OSPF ↔ EIGRP redistribution gateway
```

// CORE COMPONENTS

# The hardware stack

Five routed segments plus a dual-gateway interplanetary backbone, all stitched together with EIGRP and OSPF.



01

## RESEARCH

10.0.10.1/24

Scientific compute LAN, ACL-restricted access to selected zones.

02

## MEDICAL

10.0.20.1/24

Critical care bay with broad intra-habitat reach.

03

## HABITAT

Router-on-a-Stick

Inter-VLAN routing across 4 segmented habitat VLANs.

04

## SECURITY

10.0.200.1/24

Hardened zone, inbound ACL filtering all return traffic.

05

## GATEWAYS

Mars / Earth

Dual routers performing NAT/PAT and protocol redistribution.



// ROUTING PROTOCOLS

# EIGRP meets OSPF at the gateway

// EIGRP - Internal Mars

// OSPF - Earth Uplink

## Enhanced Interior Gateway

All Mars-side routers run EIGRP AS 10 with auto-summary disabled, providing fast convergence and a single, unified internal routing domain.

```
router eigrp 10
 network 10.0.0.0
 no auto-summary
!
redistribute ospf 1
 metric 10000 100 255 1 1500
```

## Open Shortest Path First

The interplanetary segment between Mars and Earth gateways runs OSPF Area 0. Mutual redistribution with EIGRP allows end-to-end reachability.

```
router ospf 1
 network 200.0.10.0 0.0.0.3 area 0
 redistribute eigrp 1 subnets
!
router eigrp 1
 network 10.0.4.0 0.0.0.3
```

// VLAN SEGMENTATION

# Habitat **zones**, isolated by design

**10****Residential**

Habitat living quarters

10.0.110.0/24

**20****Greenhouse**

Food production systems

10.0.120.0/24

**30****Robot Maintenance**

Autonomous unit servicing

10.0.130.0/24

**40****Tourist**

Visitor &amp; expedition traffic

10.0.140.0/24

// HABITAT ROUTER - SUBINTERFACE TRUNK

```
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 10.0.110.1 255.255.255.0
 ip helper-address 10.0.100.2
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 10.0.120.1 255.255.255.0
 ip helper-address 10.0.100.2
!
/* trunk on switch fa0/1 */
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40
```

## // APPLICATION SERVICES

# Application-layer infrastructure

A dedicated server farm (10.0.100.0/24) hosts every habitat-wide service, with DHCP helpers bridging requests from each VLAN and segment.

01

**DHCP**

10.0.100.2

Centralized IP assignment across all VLANs and LANs via ip helper-address relay.

02

**DNS**

10.0.100.3

Resolves mail.lab.com, dhcp.lab.local, and internal habitat records.

03

**SMTP / POP3**

10.0.100.4

Inter-department email service with pc0@lab.com and pc1@lab.com accounts.

04

**HTTP / WEB**

10.0.100.x

Habitat intranet portal accessible from permitted VLANs only.

# Hiding Mars behind a **single** public IP



## Why NAT overload?

Mars uses private 10.0.0.0/8 addressing internally. To conserve the scarce public IP space allocated to the colony, all outbound traffic is translated through a single IP – 200.0.10.1 – on the Mars gateway.

- ▶ Inside / outside interfaces defined per gateway
- ▶ ACL1 permits 10.0.0.0/8 for translation
- ▶ Established TCP & echo-reply allowed inbound
- ▶ Unsolicited Earth→Mars sessions dropped by design

// ACL POLICY MATRIX

# Least-privilege access matrix

Extended ACLs applied inbound on each subinterface and segment interface enforce who-can-talk-to-whom across the habitat.

| DEPARTMENT   | PERMITTED DESTINATIONS                     |
|--------------|--------------------------------------------|
| Research     | Medical, Robot Maintenance                 |
| Medical      | Research, Residential, Greenhouse, Tourist |
| Residential  | Medical, Tourist                           |
| Greenhouse   | Medical                                    |
| Robot Maint. | Research                                   |
| Tourist      | Medical, Residential                       |
| Security     | All departments (monitoring)               |

// CHALLENGES SOLVED

# From **breakage** to stable link

10

01

## Earth PC couldn't reach Mars PCs

PAT hid internal 10.x addresses, breaking inbound sessions from Elon's PC.

### SOLUTION

Identified that PAT by design drops unsolicited inbound traffic. Documented as expected behaviour; ACLs now allow only established/echo-reply return paths.

02

## EIGRP↔OSPF routes not propagating

Mars gateway ran both protocols, but routes weren't being shared between domains.

### SOLUTION

Added mutual redistribution with proper seed metrics (10000 100 255 1 1500) and "subnets" keyword on OSPF redistribution.

03

## DHCP requests failing across VLANs

VLAN-segmented PCs couldn't obtain IPs from the central DHCP server.

### SOLUTION

Configured ip helper-address 10.0.100.2 on every router subinterface and LAN interface to relay broadcast DHCPDiscover messages.

04

## Security zone being scanned by departments

Internal segments were initiating pings and unwanted connections to the security LAN.

### SOLUTION

Implemented SEC\_INBOUND ACL on Gig0/0 out — denying all 10.0.0.0/8 → 10.0.200.0/24 echo, while permitting established TCP and DNS replies.

// TRANSMISSION END

# Mars is online.

A fully segmented, secured, and interplanetary-capable network — designed, configured, and verified end-to-end. Every packet has a purpose, every VLAN a policy, every route a reason.

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